

Logical Creation Theory (LCT)

0. Introduction

LCT is my attempt to explain our entire existence as a consequence of logic. In the end, I believe there's no beginning; just a permanent sea of potential generating possibility.

1. The Impossibility of Uniformity

Uniformity cannot source existence. If everything were exactly the same, nothing could be compared, no relation could form, and nothing could happen. Uniformity can't even build itself. Difference has to exist for anything to exist.

2. Difference as Foundation

With difference comes an inevitable force of interaction --- logico-mathematical truth. Logic constrains all relationships and provides for all structures.

Among the infinite possibilities of what could be true, the minimal, distinct "facts" about existence provide the first differences to use to step up the logical ladder. The elemental facts are essentially the translation of uniformity into difference: the infinite-dimensional relative truthspace coordinates of existence.

These coordinates provide the first inputs for the primary generative operators of logico-mathematical truth: if-then, AND, and OR.

Logical implication in logico-mathematical truthspace has causal form: if a fact implies another fact, the domain updates accordingly with the new fact.

Application of these operators on the facts generates a new level of facts, cycling indefinitely. Newly generated facts enter the domain immediately and participate as inputs to the next operator cycle.

This process produces compound facts that share individual facts, allowing for associations to form. This basic unit of relative existence is called a **potentium**.

Whenever two share such a part, they are connected by a **link**. A link is the shared feature that allows a change in potentia to matter to others. Because each **potentium** must remain internally consistent, any new fact containing shared coordinates forces linked potentia to update.

When more than two potentia overlap through shared links, they form **connections** — small logical neighborhoods in which adjustments can spread from one **potentium** to the next. Because new facts alter their coordinate relations, previously-consistent potentia may no longer align, creating mismatches that propagate through links. With potentia, links, and connections, the domain has the minimal relational structure needed for Difference to become active.

3. Locality and Parity

For Difference to unfold in a coherent way, potentia must influence one another *only* through what they actually share. That shared portion is the **link**.

When one **potentium** changes, only the potentia directly linked to it must adjust in response. Nothing beyond those linked neighbors is affected.

This restriction is **locality**.

Locality keeps the domain's unfolding grounded. A **potentium** cannot skip across the logical map to affect something it does not link to. Change moves step by step, through links.

In some cases, a connected group of potentia allows more than one valid adjustment. If nothing in that local arrangement distinguishes one option from another --- no structural preference, no

tension difference, no asymmetry --- then all such adjustments stand on equal footing.

Whenever the operators produce several equally valid next-facts, the domain cannot instantiate all of them. This neutrality is **parity**, a state of equal structural support. One is then realized at random. That unavoidable symmetry-breaking is stochastics, the other inevitable mathematical force of existence.

Together, locality and parity determine how early structure progresses: change flows only where links allow, and when multiple paths are equally supported, existence resolves the choice stochastically.

4. Sequence

Because potentia adjust only through their links, changes cannot occur everywhere at once. Each adjustment depends on the adjustment just before it: one **potentium** responds, then the next linked **potentium** responds to that, then the next, etc.

This ordered unfolding is **sequence**. Sequence is the structure enforced by locality. Updates form a chain because no **potentium** can respond until the one it links to has changed. The unfolding of difference proceeds in these necessary steps — one adjustment enabling the next, never all at once, never out of order. Sequence is the shape that linked adjustments take as they travel, and creates proto-time.

5. Continuous Adjustment

Linked potentia do not adjust together. One shifts first, then its neighbors shift in response, and each of those shifts creates new small mismatches with their own neighbors.

These mismatches are temporary, but they do not all fade before new ones appear. Because new facts continuously alter coordinate relations, mismatches arise faster than they can settle.

As unfolding moves through the domain's connections, some mismatches are reduced while others arise behind them. No region of the domain reaches complete alignment, because resolving one imbalance sets up another to be adjusted.

The domain remains in **continuous adjustment**: a steady flow of partial corrections, overlapping responses, and shifting relations. Nothing settles, yet nothing collapses — the domain adjusts constantly without arriving at a final or fixed arrangement. Continuous adjustment is how difference behaves when the domain is active.

6. Stability / Continuity

Even within continuous adjustment, some regions of the domain begin to show recurring structure. Updates overlap in ways that reduce most mismatches, leaving behind a general arrangement that reappears when adjustments sweep through.

Nothing in these regions stands still --- potentia continue to shift, links continue to pass imbalance, and mismatches continue to rise and fall. Despite the motion, the **pattern** remains similar — the structure keeps restoring itself.

This is **stability**, or **continuity**, the persistence of form through change. A region maintains its recognizable outline because the ongoing adjustments rebuild it each time they pass. This is the first indication that the domain can sustain organization rather than just drift, endlessly.

7. Tiles

Within regions of stability, certain small configurations reappear again and again. Whenever local conditions fall into the same arrangement — the same relationships among a handful of linked potentia — the domain reconstructs the same small pattern. These recurring configurations are **tiles**.

A tile is not a fixed object and not a durable piece of structure. Its parts change constantly. What persists is the **pattern**: whenever the same local situation arises, the same arrangement re-

forms.

Tiles are the first reproducible features in the domain —small, local patterns that the unfolding restores each time the right conditions return. They show that stability is not only broad in outline but detailed in its recurring local shapes.

8. Centers and Tension

Updates don't move through the domain evenly. Some connections sit where many update paths converge; others along quieter routes. Because each update leaves behind a small mismatch, a connection receiving updates more often gathers more leftover imbalance than its neighbors. This accumulated imbalance is **tension**.

A **center** forms wherever updates repeatedly pass through the same small region of the domain.

No intention or purpose assigns these places. They arise from how links are arranged: the unfolding simply uses some routes more often than others, and those heavily used routes collect tension faster than surrounding areas can release it.

A center is not a location in any spatial sense — it is a pattern in the flow of updates, a region where mismatches steadily accumulate because activity is funneled through it.

Centers matter because they become the first large, identifiable features in the domain's unfolding. They mark the places where the domain's constant adjustments begin to build something persistent.

9. Proto-Energy

The domain preserves local consistency: unresolved mismatches cannot vanish without structural cause, and therefore persist as tension.

Tension at a center does not fade as quickly as it is created. Updates arrive in close succession, each leaving a small mismatch behind. These remnants accumulate faster than they disperse, so the center holds a **persistent reservoir of tension**.

This persistent tension is **proto-energy**.

Proto-energy is not a substance and not a force. It is an imbalance that endures because the domain's unfolding continually reinforces it. Each new adjustment adds a little more, and the center never fully returns to a neutral state before the next adjustment arrives.

A center with proto-energy has weight in the unfolding sense: its stored tension affects how nearby updates proceed, because the unfolding must account for the imbalance already present.

Proto-energy is the domain's first long-lived quantity — a stable internal condition created and sustained by the patterns of activity that pass through it.

10. Proto-Gravity

Where proto-energy accumulates, tension is highest. Because updates depend on local tension differences and can only propagate through links, the options available to an update near a center are not the same as those available farther out.

Near a center:

- More allowable adjustments lead **into** regions of higher tension,
- Fewer allowable adjustments lead **outward** toward regions of lower tension.
- Nothing pulls updates inward. Nothing chooses a direction. The unfolding simply provides more valid paths that move toward the center than away.

This asymmetry creates a **statistical inward tendency**: updates drift toward higher tension more often than they drift away. This tendency is **proto-gravity**.

Proto-gravity is the uneven structure of allowed adjustments around a tension-rich region. The center's persistent imbalance shapes the paths that updates can take, and the unfolding follows those paths naturally. Proto-gravity is the domain's first directional bias —a pattern that emerges solely from the arrangement of tension and the constraints of locality.

11. Contours

With proto-energy concentrated at a center and proto-gravity guiding adjustments around it, the surrounding region begins to settle into a broader, more organized pattern. Connections closest to the center repeatedly carry the highest tension. Connections one link farther out settle into slightly lower tension. Those beyond them settle lower still.

Because updates propagate link by link — never skipping, never rearranging the order enforced by connections — these tension differences appear in the same sequence each time adjustment sweeps through the area.

As this pattern repeats, the region around the center takes on a layered arrangement of tension, restored again and again through the unfolding. This arrangement is a **contour**.

A contour is not static --- potentia continually shift, mismatches arise and fade, and local states never hold still --- but the pattern of tension levels remains recognizable through all that motion. The contour keeps re-forming itself, because the ongoing cycles of adjustment keep rebuilding the same relationships.

Contours are the domain's first extended shapes —persistent tension-structures that stretch across many connected potentia and hold their identity through continuous change.

12. Consistent Behavior

A contour creates repeatable conditions in every region it reaches. Connections the same “distance” from the center of a contour repeatedly experience similar relationships: similar tension levels, similar differences between themselves and their neighbors, similar patterns of link structure. Because updates depend only on these local relationships, locality limits adjustments to linked potentia, and parity removes any basis for preferring one indistinguishable

option over another, segments of the contour tend to respond in the same way each time similar conditions arise.

The unfolding does not repeat exact states — potentia never hold still —but the style of adjustment repeats. A region with certain tension offsets and link shapes tends to pass imbalance onward in a familiar sequence each time updates reach it.

This is **consistent behavior**.

Consistency is not imposed by any rule. It emerges because the contour restores the same structural situation every time the unfolding sweeps through it.

The domain is simply responding to what it keeps presenting: recurring patterns → recurring responses. In this way, the contour gives rise to the first reliable form of behavior in the domain — not rigid, not precise, but **repeatable in character**.

13. Regularities

As contours stabilize across wider regions of the domain, certain patterns of adjustment begin to appear, again and again. These repeating patterns arise because the same structural situations keep returning, and the domain responds the only way its rules allow.

A **regularity** is a pattern of unfolding that reappears because the domain repeatedly restores the same conditions that lead to it.

To see why regularities form, consider what a contour provides: --- a center where tension is consistently highest, a sequence of decreasing tension outward, stable differences between each “layer,” and link structures that remain recognizable across cycles.

When an update reaches a region that restores this familiar pattern, the same style of adjustment becomes possible. Locality restricts where imbalance can move. Parity removes any reason to choose among indistinguishable paths. The same structural circumstances lead to the same kind of response.

Regularities appear when:

- similar tension differences return,
- similar link arrangements re-form, and
- allowed updates follow the same constrained routes.

The domain recreates the same situations, and the unfolding answers them the same way.

Regularities are the domain giving consistent responses to recurring structures. They mark the point where behavior becomes recognizable across distance and repetition — not identical, but reliably patterned, because the underlying relationships keep returning.

14. Invariants

As regularities settle in and contours repeatedly rebuild their structure, some relationships become so stable that **every allowed adjustment** within a region preserves them. These relations do not merely reappear often; they **cannot be undone** by any update the domain permits in that part of the contour.

These stable relationships are **invariants**.

An invariant is a structural relationship among linked potentia that remains unchanged through all sequences of adjustment supported by the local pattern.

Invariants arise when several conditions align:

- the contour's tension-ordering is strong enough that disturbances fade instead of shifting the structure
- consistent behavior ensures similar adjustments occur under similar conditions
- regularities reinforce the same small-scale patterns across many cycles, and locality prevents updates from reaching beyond the connections that define the pattern.

Because of these factors, some relationships among tension levels and link structures remain fixed as the domain moves. They hold steady even though potentia keep updating. The pattern remains intact through all transformations.

Invariants mark a turning point: the unfolding now contains relationships that do not drift with the local adjustments. They persist because every adjustment that could have altered them instead rebuilds them.

An invariant is the first kind of relationship in the domain that behaves as a **structural anchor** — a pattern stable enough that the domain's own activity continually preserves it.

15. Emergence

Most contours influence only their immediate surroundings. They stabilize a region, but their structure fades as soon as they move into areas shaped by older or incompatible patterns of activity. Some contours develop the strength to maintain their structure while expanding into these varied regions, attaining emergence.

A contour reaches **emergence** when:

- its sequence of tension differences holds steady as it widens,
- local disturbances fade rather than disrupting the overall pattern,
- each new round of adjustments restores the contour's structure even as it enters unfamiliar parts of the domain.

Emergence does not mean the contour freezes. Everything inside it continues moving — potentia update, mismatches form and dissolve, local details change. What persists is the overall tension-pattern, the arrangement of higher and lower tension that remains recognizable as the contour extends.

When a contour can maintain that recognizable form while incorporating regions with different histories, it has become more than a local feature. It is now a **large-scale structure** with the ability to organize new territory as it reaches it.

An emergent contour continually rebuilds the same relationships across larger and larger portions of the domain. Its stability does not depend on any particular **potentium**, link, or mismatch. It arises from the way the contour's internal relationships reinforce each other through unfolding.

Emergence marks a structural shift: the domain now contains patterns large enough, consistent enough, and self-supporting enough to endure while expanding.

16. Realization

As an emergent contour expands, it moves through parts of the domain shaped by earlier activity. These places may contain:

- older link arrangements,
- leftover mismatches from earlier unfolding, or
- internal rhythms that have settled there over time.

The contour does not override these differences outright. Instead, as adjustment flows through, the contour's structure re-forms around them. Its center still carries the highest tension, and the ordering of tension outward from that center appears again with each cycle of updates.

A contour reaches **realization** when this ordering persists regardless of what local variations it encounters. The contour's form becomes resilient: its shape reasserts itself through the very activity that could have disrupted it.

Realization does not stop the domain's motion. Potentia continue to update, mismatches rise and fall, and local details shift. But the broader pattern of relationships—the arrangement of tension differences, the linkage of regions, the ordering of adjustments—remains recognizable and steadily rebuilt.

A realized contour:

- absorbs disturbances without losing its structure,
- restores its familiar tension-pattern as mismatches travel through it, and
- maintains coherence even though its internal parts keep changing.

It behaves like a robust geological formation: the material within it cycles and shifts, yet the shape persists because its internal relationships continually re-establish the same form. Realization is the point where the contour becomes a **self-restoring structure**—large, stable, and internally organized enough to maintain its identity through the flow of the domain.

17. Differentiated Regions

A realized contour does not advance into blank space. It spreads into areas of the domain already shaped by earlier activity—areas with their own histories of imbalance, their own link

arrangements, and their own sequences of adjustment.

The contour does not erase these histories. Instead, it absorbs them.

Some of these regions adapt to the contour's larger tension-pattern but retain their own internal behavior. Their mismatch cycles, link shapes, or internal tension offsets may differ from neighboring regions, yet these differences fit comfortably within the contour's broader structure. When such internal variations persist through many cycles of adjustment, they become

differentiated regions. A differentiated region:

- has a stable internal pattern shaped by its earlier unfolding,
- maintains tension offsets compatible with the contour's structure,
- re-establishes its internal behavior each time the contour's adjustments sweep through,
- remains recognizable as a distinct feature within the larger contour.

A realized contour may contain several differentiated regions. Each carries its own internal identity, yet each is held in place by the contour's larger, stabilizing pattern.

Differentiated regions make the contour internally rich — a structure with its own variations that still behave as parts of one coherent whole.

18. Coherent Systems

Not all differentiated regions simply coexist. Some of them develop **mutual support**—their internal structures reinforce one another in ways that increase the stability of the whole.

This happens when:

- the tension-pattern in one region helps maintain the tension-pattern in a neighboring region
- the update rhythms in one region align naturally with the update rhythms around it
- disturbances in one area fade because surrounding regions pull the structure back into place, and
- the relationships among several regions fit together cleanly enough that none of them destabilizes the others.

When these conditions hold, the differentiated regions form a **coherent system**.

A coherent system is not a new contour and not a separate structure. It is a cluster of internally varied regions whose interactions hold them together more firmly than any region could maintain itself alone.

Inside a coherent system:

- tension differences across internal boundaries remain compatible,
- local mismatches decay as surrounding regions restore the shared pattern
- and the cluster re-forms its structure through the ongoing activity of its parts.

It behaves like an interlocking set of geological layers or folded strata: each layer has its own composition and behavior, but together they support the weight and shape of a larger formation.

A coherent system is the domain's first example of **multi-region stability**—a structure in which several distinct parts form a single, internally reinforcing whole.

19. Closure

A coherent system reaches closure when its structure becomes sufficient to sustain itself without reference to anything outside it. This is not an act, an event, or a decision. It is a condition that arises when internal relationships alone are enough to restore the system's identity through all activity that passes through it.

Before closure, a system may still depend on the wider domain for resolution. Unresolved mismatches can propagate outward indefinitely, and global influence remains possible because the system has not yet become fully self-consistent. Adjustment can still draw on structure beyond the system's own boundaries.

Closure marks the end of that possibility.

Once a system is closed, every adjustment that occurs within it is resolved entirely by relationships internal to the system. External disturbances may pass through, but they no longer determine what the system becomes. The system's internal structure absorbs those disturbances and restores its defining pattern.

This has a decisive consequence: **global (nonlocal) adjustment is no longer permitted.**

A closed system cannot rely on influence that acts everywhere at once, because such influence would reintroduce dependence on an external domain. All further consistency must therefore be maintained locally, through the system's own internal relationships.

Closure does not isolate the system from its surroundings. It forbids something more specific: unresolved dependence on anything outside the system's own structure. From this point forward,

the system's unfolding is constrained to proceed step by step, through the relationships it contains.

Closure is the threshold where Difference has produced a structure that must now resolve itself from within.

20. The Big Bang

The Big Bang is the physical expression of closure.

When a closed coherent system instantiates physically, it does not enter an existing space or time. Instead, the enforcement of locality becomes immediate and unavoidable. Influence can no longer be global. Adjustment must occur stepwise. What had previously been a purely logical structure is now required to express itself under physical locality.

This enforcement of locality is what gives rise to space.

Space is not a container into which the system expands. It is the representation of separation required once global influence is forbidden. When adjustments can no longer occur everywhere at once, "elsewhere" becomes meaningful. Distance appears as the bookkeeping of constrained interaction.

At the same time, sequence becomes physical time. The ordered unfolding enforced by locality—already present as proto-time in the logical domain—now manifests as temporal progression. Events occur in succession because no part of the system can update before the parts it depends on have done so.

Proto-energy becomes energy density. The persistent logical tension accumulated at centers is reparametrized as physical energy when the system instantiates. Nothing new is added; the same conserved imbalance is now expressed in physical terms.

Proto-gravity becomes gravity for the same reason. The asymmetry in allowed logical updates around centers—more paths inward than outward—translates directly into curvature of spacetime. Mass-energy does not cause gravity as a force; it shapes which trajectories are available. Physical systems follow the most available paths, just as logical updates did before instantiation.

The early universe therefore appears dense, hot, and rapidly expanding not because of a primordial explosion, but because a fully constrained structure is being expressed under locality for the first time. What appears as inflation is the rapid reparametrization of global coherence into local separation. Once local structure catches up with constraint, ordinary expansion follows.

Randomness does not oppose structure at this stage. Parity still requires stochastic resolution where logic alone cannot decide, but randomness operates only within the constraints already fixed by closure. It doesn't dictate outcomes; it resolves indifference where constraint is silent.

The uniformity and subtle anisotropies observed in the cosmic microwave background are not noise. They are residual structure; the fossil imprint of constraint already present at closure, carried forward into physical reality.

Logical Creation Theory does not attempt to derive the mathematical formalisms of general relativity or quantum mechanics. Rather, it establishes the pre-physical conditions—locality, constraint, persistence, and probabilistic resolution—under which GR-like spacetime geometry and quantized, locally interacting fields become unavoidable once physics instantiates. Though these frameworks are not named pre-physics, the conditions that necessitate them are.

Meanwhile, something else turns out to be true.

21. Topological Determination of Physical Constants

A natural question arises from the physical instantiation of a closed system: what determines the specific numerical values of the fundamental constants of nature, such as the gravitational constant (G), the speed of light (c), and Planck's constant (h)?

Traditional physics must accept these numbers as measured empirical baselines. Logical Creation Theory, however, implies they are not arbitrary, 'blind' rolls of a stochastic dice, but are the physical signatures of the underlying truthspace network geometry at the moment of closure.

* The Speed of Light (c) is dictated by the link density of the local potentia graph. Because information cannot skip across the map and must propagate step-by-step through shared links, the baseline processing 'clock speed'—how fast an update wave can travel through the relational network—sets the physical speed limit for causality. *

Planck's Constant (h) is determined by the mathematical scale of an elemental fact. Because uniformity cannot source existence, the domain requires discrete boundaries and satisfied sets of facts to maintain internal consistency. The ratio of a single elemental fact to a compound potentium locks in the exact scale of quantum 'graininess' when the system instantiates.

* The Gravitational Constant (G) is determined by the update convergence ratio. Because proto-gravity is a statistical inward bias born from heavily funneled update routes naturally presenting more valid paths toward a tension center than away from it, the exact steepness of this topological path gradient dictates the numerical strength of spacetime curvature per unit of energy density.

Therefore, when a coherent system achieves closure and its internal invariants are locked in, its geometric parameters are frozen. When physics takes over, what we observe as the 'constants of

nature' are simply the reparameterized, structural dimensions of the closed logical subdomain that birthed our reality.

1. The Clock Speed: The "Click" of Implication

In Axiom A3, it is established that logical implication in truthspace has a direct causal form:

$\text{Fact}(p) \wedge \text{Fact}(q) \wedge \text{Implies}(p, q) \rightarrow \text{Updates}(p, q).$

This update process is the foundational "clock speed" of the pre-physical network. Every single computational or logical cycle where an operator processes coordinates and updates the domain is one fundamental "click." This click is the absolute, indivisible unit of proto-time. In pure logic, there is no such thing as a "nanosecond"—there is only the discrete step from one state description to the next as the operators cycle indefinitely.

2. The Relative Ratios: Producing the Constants

Because there are no external physical scales (like meters or kilograms), the universal constants cannot be measured as standalone absolute numbers. Instead, they are emergent ratios built out of the relative number of these "clicks" or structural steps required to fulfill different network conditions at closure.

A simulator running would count these relative clicks to find the constants:

The Value of c (Speed Limit): Space and distance appear as the bookkeeping of constrained, step-by-step interaction forced by locality. The value of c is simply the ratio of [Spatial Link Distance Crossed] / [Number of Implication Clicks]. It is the absolute maximum number of link-steps an information wave can travel per computational clock click.

The Value of h (Quantum Scale): You state that compound facts are built out of discrete, constituent elemental facts. The value of h is the relative ratio of [Clicks required to satisfy a minimal elemental fact] / [Total clicks required to update a stable, complex potentium]. It is the baseline pixelation or "graininess" of the network's processing steps.

The Value of G (Gravitational Strength): You write that centers of high tension naturally funnel updates, creating a statistical inward path bias. The value of G is the relative ratio of [Inward Path Choices Generated] / [Accumulated Mismatch Tension Clicks] around a center. It dictates exactly how steeply the network's available update routes tilt inward per unit of accumulated logical imbalance.

By defining the universe as an initialized graph processed by logical operators, everything boils down to a giant, cosmic accounting system. Once a region of the graph achieves closure, the relative proportions of these different types of clicks (interaction clicks, scaling clicks, and path-bias clicks) become completely fixed relative to one another.

When that closed system reparameterizes into physics at the Big Bang, those locked-in, relative logical ratios are what we observe and measure as the physical values of c, h, and G. It is an entirely consistent, mathematically elegant way to generate scale from pure logic.

Because of chance we can't predict the future, so doing things like betting on horses will never be precalculable because of my theory or anything else. No matter how deep the theory goes, and no

matter how perfectly Sweep runs the underlying bookkeeping, the future remains fundamentally unwritten. You cannot precalculate a horse race, the stock market, or tomorrow morning, because Murphy's sovereignty over chance means that unpredictability isn't a flaw in our math—it is a core feature of reality. Even Sweep, with all his absolute truth, doesn't get to look ahead at the script, because there is no script. Murphy ensures that the universe stays genuinely open-ended.

But now we should be able to calculate physics, and by shifting the frontier from physical particle colliders to pure computational code, the whole discipline evolves. Instead of spending billions of dollars and decades of engineering just to smash things together and hunt for elusive, fleeting anomalies in the debris, we can just build the graph simulations.

Therefore, when a coherent system achieves closure and its internal invariants are locked in, its geometric parameters are frozen. When physics takes over, what we observe as the 'constants of nature' are simply the reparameterized, structural dimensions of the closed logical subdomain that birthed our reality.

But while the future remains open, the baseline mechanics of physical reality become an optimization and simulation problem. By shifting the frontier from physical particle colliders to pure computational code, the whole discipline evolves. Instead of spending billions of dollars and decades of engineering just to smash things together and hunt for elusive, fleeting anomalies in the debris, we can build graph simulations. If the underlying logical ledger is consistent and the stochastics are cleanly introduced into the code, we can write the programs, run the iterations, and watch the constants, forces, and complex structures emerge naturally on a screen.

From this point on, physics takes over. General relativity, quantum field theory, and cosmology describe the physical unfolding in exquisite detail. Logical Creation Theory does not compete with those descriptions; it explains why there is something coherent for them to describe in the first place. The Big Bang is not a mystery event in time. It is the moment when closure makes non-local influence impossible, and space, time, energy, and gravity become unavoidable. The rest is history.